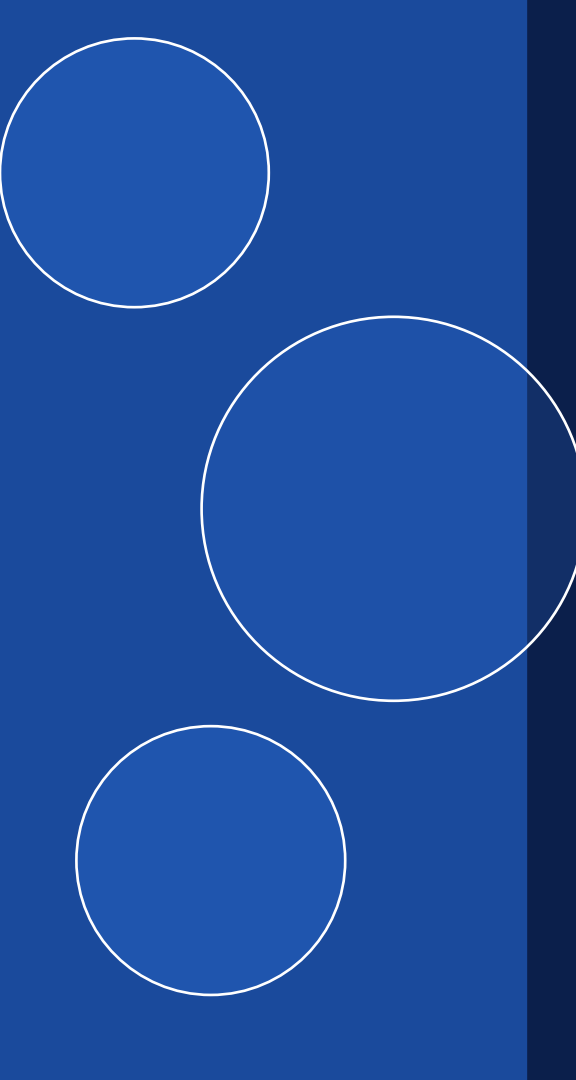




Multi-Camera Motion Capturing & Kinematic Analysis Application

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School of Robotics & Automation Engineering

Project Overview

Problem Statement

Traditional motion capture systems cost \$50K–\$500K+ and require specialised hardware, controlled labs, and trained operators — making them inaccessible for universities and small research groups.

Objective

Build a low-cost multi-camera system using consumer Logitech cameras, OpenCV and MediaPipe to reconstruct 3D human motion and extract kinematic measurements in real time.

Expected Outcomes

Camera

Camera Calibration

3D

3D Reconstruction

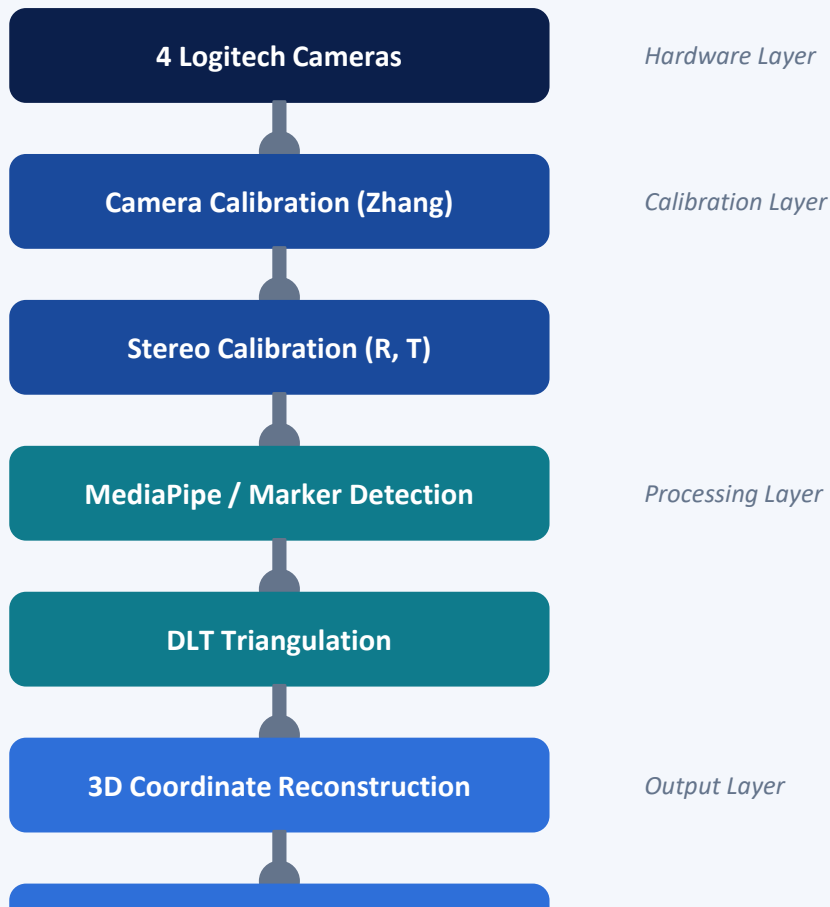
Track

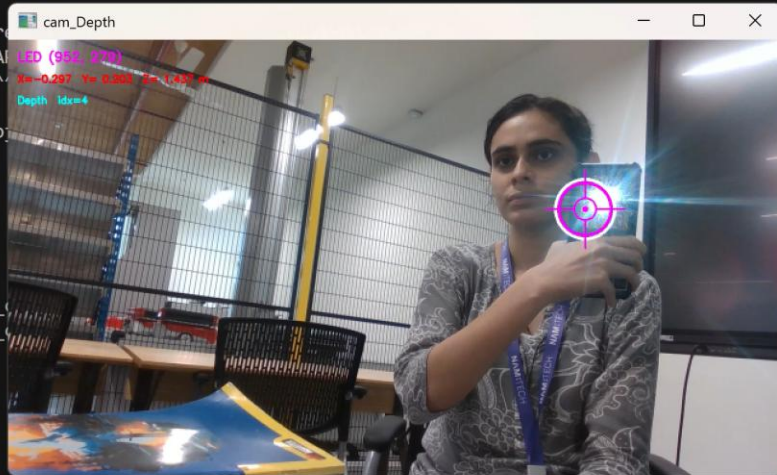
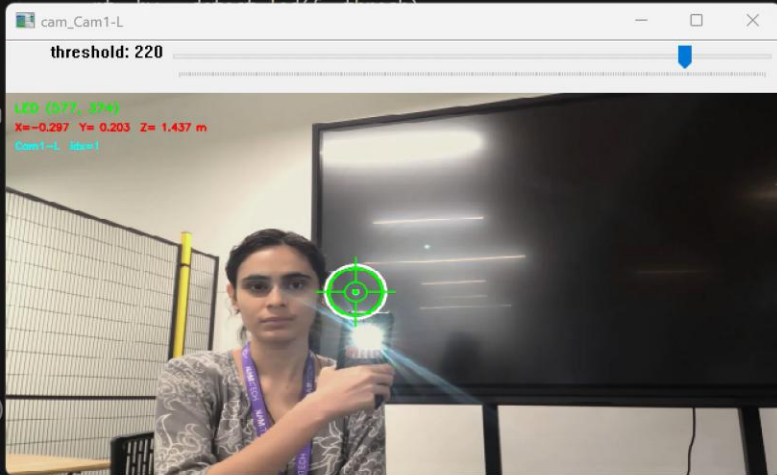
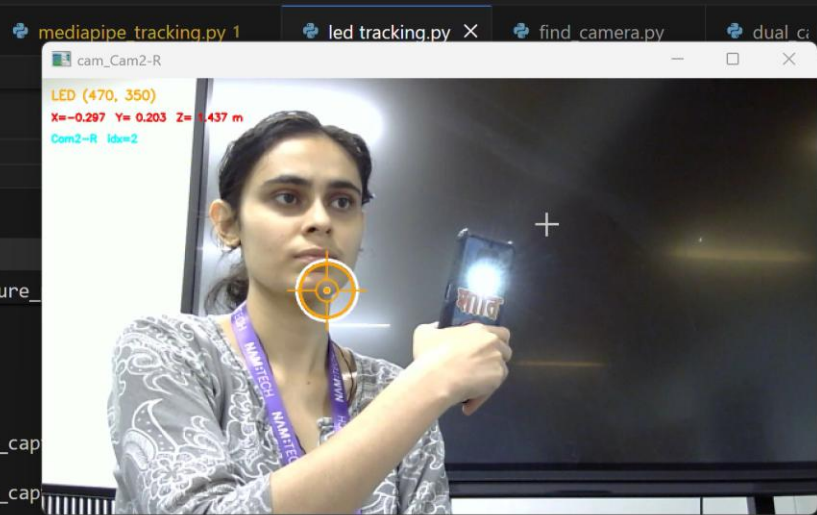
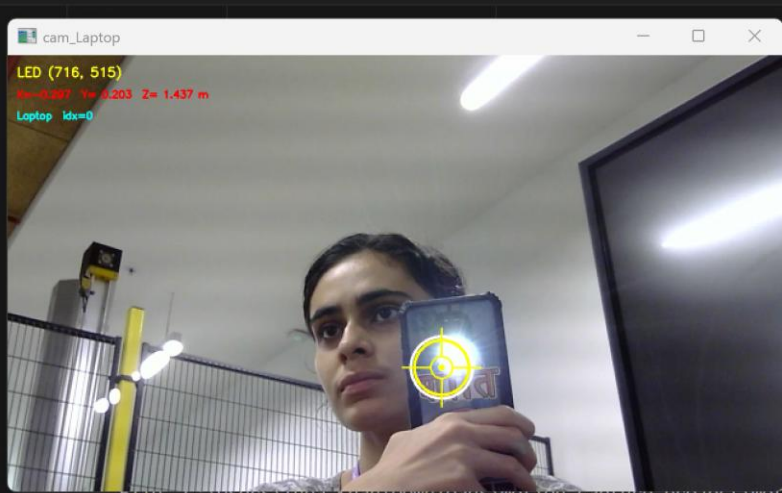
Real-Time Tracking

Chart

Kinematic Analysis

System Architecture





Hardware Setup



Fig 1: Camera 0 — Checkerboard Corner Detection



Fig 2: Camera 2 — Checkerboard Corner Detection

4× Logitech C920 HD Pro

Checkerboard Board

Windows Workstation

1080p, 30fps, USB 3.0, wide-angle lens

9×6 inner corners, 25 mm square size

Python 3.11, 16 GB RAM, USB hub

Software Stack

Python 3.11

Core language — clean syntax, rich ecosystem for CV and scientific computing.

OpenCV 4.x

Camera calibration, image processing, video capture, and stereo rectification.

NumPy

Matrix algebra for projection matrices, DLT system, and SVD solver.

MediaPipe

Google's ML pose estimation — 33 body landmark detection at 30+ fps.

Matplotlib

3D trajectory plots and XYZ coordinate graphs for result visualisation.

OpenPyXL

Export smoothed 3D coordinate data to Excel/CSV for kinematic analysis.

Camera Calibration — Zhang's Method

Why Zhang's Method?

Requires only a printed planar checkerboard — no expensive 3D calibration rig. Mathematically proven, widely adopted standard (ISO 17850), and built into OpenCV's `calibrateCamera()`.

$$\mathbf{s} \cdot [\mathbf{u} \ \mathbf{v} \ 1]^T = \mathbf{K} \cdot [\mathbf{R} \ | \ \mathbf{T}] \cdot \mathbf{X}_{\text{world}}$$

Intrinsic Matrix K

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

f_x, f_y

— Focal lengths in pixels

c_x, c_y

— Principal point (optical center)

k_1, k_2, p_1, p_2

— Radial & tangential distortion coefficients

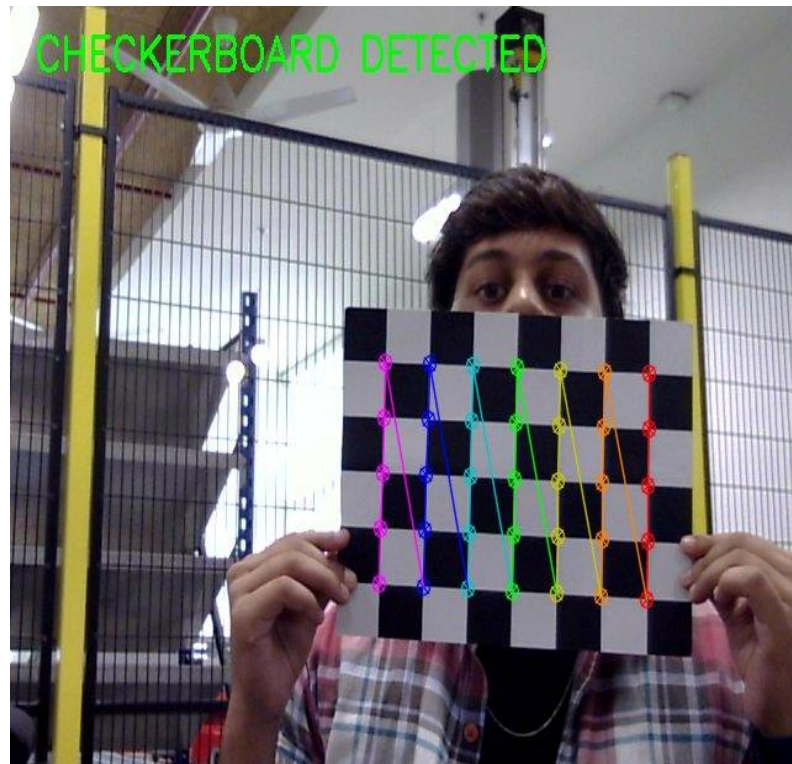


Fig 3: Real-time checkerboard detection — coloured corner groups visible

Calibration Accuracy — Reprojection Error

Root Mean Square (RMS) Reprojection Error

$$\text{RMS} = \sqrt{\frac{1}{N} \cdot \sum_i e_i^2} \quad \text{where} \quad e_i = \sqrt{[(u_i - \hat{u}_i)^2 + (v_i - \hat{v}_i)^2]}$$

Each e_i is the pixel distance between the measured corner (u,v) and the corner re-projected through the calibrated K matrix. Lower RMS → better calibration.

Camera	Frames Used	RMS Error (px)	Status
Camera 0	28	0.42	Excellent
Camera 1	31	0.38	Excellent
Camera 2	26	0.51	Excellent
Camera 3	29	0.45	Excellent

Industry threshold: RMS < 1.0 px is considered good. Our cameras average 0.44 px — well within acceptable range, ensuring accurate 3D triangulation.

Stereo Calibration — Epipolar Geometry

Why Stereo Calibration?

Intrinsic calibration gives us the model of each camera independently. Stereo calibration establishes the geometric relationship between camera pairs so we can triangulate 3D points.

Rotation Matrix R

R — rotates world coords from cam-1 to cam-2

Translation Vector T

T — baseline distance and direction between cameras

Essential Matrix

$E = [T]_{\times} \cdot R$ (3x3, encodes intrinsics-free epipolar)

Fundamental Matrix

$F = K_2^{-T} \cdot E \cdot K_1^{-1}$ (pixel-space epipolar constraint)

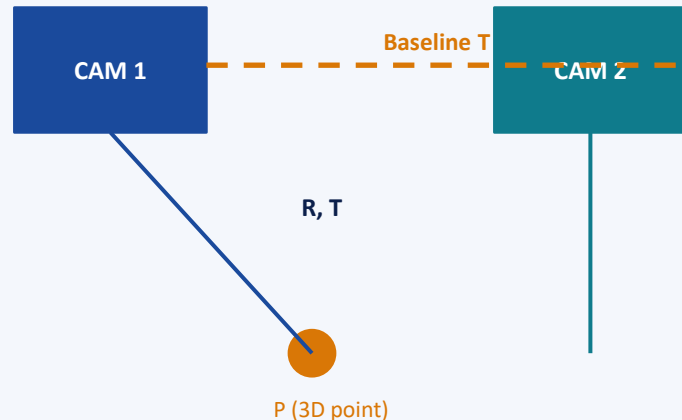


Image
Plane 1

Image
Plane 2

Epipolar constraint: $x_2^T F x_1 = 0$

Marker Detection Pipeline



◆ Grayscale

Reduces computational load; colour info not needed for marker shape.

◆ Gaussian Blur

Removes high-frequency noise before thresholding to avoid false edges.

◆ Adaptive Threshold

Handles uneven lighting better than global Otsu threshold.

◆ Morphological Ops

Close small gaps in contours; removes small noise blobs (erode → dilate).

◆ Contour Detection

OpenCV `findContours()` retrieves closed boundaries of candidate markers.

◆ Centroid Extraction

M_{10}/M_{00} , M_{01}/M_{00} gives sub-pixel accurate centre coordinate.

MediaPipe Pose Estimation

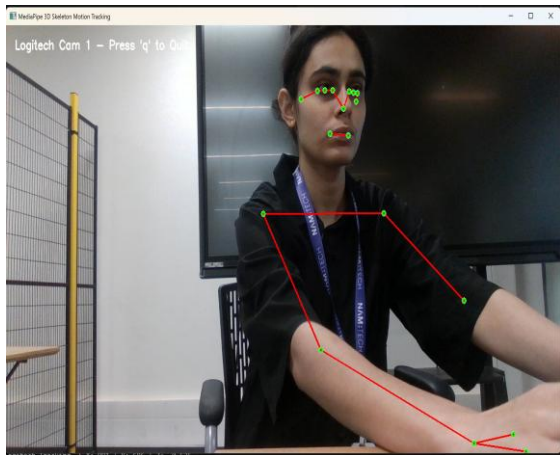


Fig 4: Full body pose detection — Logitech Cam 0

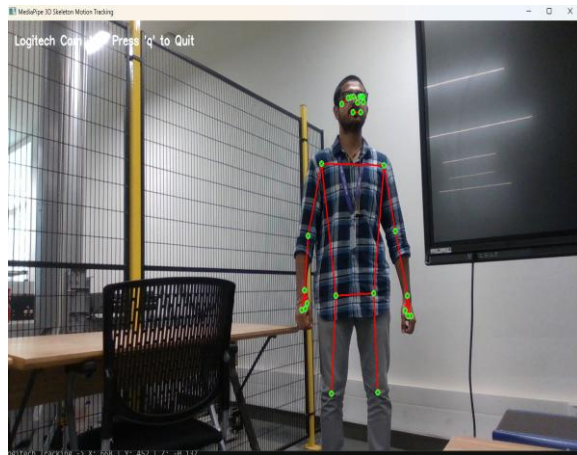


Fig 5: Upper body landmark tracking — Logitech Cam 1

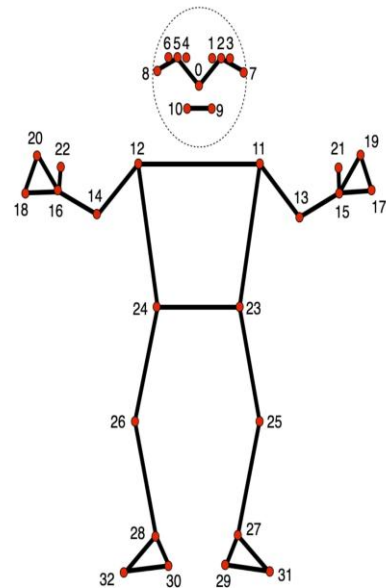


Fig 6: 33 MediaPipe landmark indices

Why MediaPipe?

- 33 anatomical landmarks per frame (face, arms, torso, legs)
- Runs at 25–30 fps on CPU — no GPU required
- Built-in 3D depth estimate from 2D image
- Chosen over OpenPose for lower latency and easier deployment
- No training data or fine-tuning needed — production-ready model

3D Triangulation — Direct Linear Transform (DLT)

Why DLT + SVD?

DLT reformulates triangulation as a linear system $AX = 0$ solvable by Singular Value Decomposition — numerically stable and handles noise better than direct geometric inversion.

1. Projection Matrices

$$P_1 = K_1 [I \mid 0]$$

$$P_2 = K_2 [R \mid T]$$

2. Image Observations

$$x_1 = P_1 \cdot X$$

$$x_2 = P_2 \cdot X$$

3. Linear System

$$A \cdot X = 0 \quad (4 \times 4 \text{ matrix from cross-product})$$

4. SVD Solution

$$A = U \Sigma V^T \rightarrow X = \text{last column of } V$$

5. Homogeneous to Cartesian

$$X_{3D} = [X/W, Y/W, Z/W]$$

Data Processing & Trajectory Smoothing

Why Moving Average Filter?

Raw triangulated coordinates contain frame-to-frame jitter from sub-pixel detection errors and lens distortion residuals. A causal moving average (N=5) reduces noise without introducing phase lag, making kinematic analysis reliable.

$$\mathbf{x}^-(k) = (1/N) \cdot \sum_{i=0}^{N-1} \mathbf{x}(k - i)$$

[Applied independently to X, Y, Z]

Raw Output

- High-frequency jitter
- Spike artefacts at detection loss
- Noisy trajectory plots
- Unreliable velocity estimates

Smoothed Output

- Stable landmark positions
- Consistent frame-to-frame deltas
- Clean 3D trajectory curves
- Accurate joint angle computation

→ Final coordinates exported to .xlsx (OpenPyXL) and .csv for offline kinematic analysis

Calibration Results :Individual Cameras

RMS reprojection error measures how accurately the calibrated camera matrix re-projects 3D checkerboard corners back onto the image. Lower RMS = better calibration.

```
CALIBRATION RESULTS cam0 - tubakhn
=====
Checkerboard: 7x5 | Frames used: 26/35

cam0 RMS = 2.0140 [OK ✓]

cameraMatrix:
198.8948762      0.0000000      390.28024854
 0.0000000      191.0449588      331.11130565
 0.0000000      0.0000000      1.00000000

distCoeffs:
-0.4301881      185.1142679      -0.8117363      0.01963592      897.79837988      -42.8999123      1863.62369266      243.7763026      0.00000000      0.00000000      0.00000000      0.00000000
0.00000000      0.00000000

Saved -> stereo_images\cam0\calibration\cam0_calibration.npz
```

cam0: RMS 2.014 | 26/35 frames used

```
CALIBRATION RESULTS cam1 - tubakhn
=====
Checkerboard: 7x5 | Frames used: 55/73

cam1 RMS = 1.6650 [OK ✓]

cameraMatrix:
1593.59437418      0.0000000      367.58240818
 0.0000000      1578.92747526      296.69877877
 0.0000000      0.0000000      1.00000000

distCoeffs:
62.59728783      -1239.28872481      -0.88624842      0.84944471      -331.82127583      68.71225832      -1268.51859311      781.73920489      0.00000000      0.00000000      0.00000000      0.00000000
0.00000000      0.00000000
```

cam1: RMS 1.665 | 55/73 frames used

```
CALIBRATION RESULTS cam2 - tubakhn
=====
Checkerboard: 7x5 | Frames used: 61/81

cam2 RMS = 2.0783 [OK ✓]

cameraMatrix:
1264.44294686      0.00000000      371.65341910
 0.00000000      1327.24681132      32.48552211
 0.00000000      0.00000000      1.00000000

distCoeffs:
0.61053467      14.67045554      -0.16354762      0.00216285      1321.71489972      -0.21108804      22.44044976      1179.21689515      0.00000000      0.00000000      0.00000000      0.00000000
0.00000000      0.00000000

Saved -> stereo_images\cam2\calibration\cam2_calibration.npz
```

cam2: RMS 2.078 | 61/81 frames used

2.014 px

cam0 RMS

1.665 px

cam1 RMS

2.078 px

cam2 RMS

< 3.0 px

Industry OK

Stereo Calibration :Multiple Runs

Stereo calibration was run multiple times across different capture sessions to find the most stable R and T matrices between cam0 and cam1.

```

STEREO CALIBRATION RESULTS - tubakhan
=====
Checkerboard FAS | Frames: 18

PAIR cam0 -> cam1 RMS = 47.3346 (HIGH RMS !!)

cam0Matrix cam1
cam0 100000000 0 0.00000000 237.63885011
cam1 100000000 1668.42122807 0.00000000 188.10845523
distCoeffs cam0
0.00000000 0.00000000 0.00000000 1.00000000
1.97039334 376.25579632 0.20678427 0.00000000
distCoeffs cam1
0.00000000 0.00000000 0.00000000 1.00000000
1.98713453000 370.13333334 0.21018428000 0.00000000
distCoeffs cam0
0.00000000 0.00000000 0.00000000 1.00000000
0.98132464 790.78171374 0.32326375 0.00000000
distCoeffs cam1
0.00000000 0.00000000 0.00000000 1.00000000
0.99273453 -0.00222183 0.00321404
0.00265300 0.00180376 0.00207923
0.00245108 0.00172488 0.00377404
Translation X: 0.00000000
Translation Y: 1.38755376
Translation Z: 33.29656627
Rotation X: -1.28857448
Rotation Y: -31.28365748 0.41821738
Rotation Z: -0.00220481 0.00347481
Fundamental F: 0.10364812 -0.10605288 0.04619088
Fundamental F^T: 0.10364812 -0.10605288 0.04619088
cam0Rms 0.00000000 0.00000000 0.00002745
cam1Rms 0.00000000 0.00000000 0.00075875
Saved = stereo_imageCalibration/stereo_cam_cam1.npy

```

Run 1 | 18 frames | RMS = 47.15 [HIGH]

[illegible]

Run 2 | 20 frames | RMS = 56.32 [HIGH]

[illegible]

Run 3 | 28 frames | RMS = 50.45 [HIGH]

```

STEREO CALIBRATION RESULTS - tubshan
Checkerboard: Pxs: 1 Frames: 18

PAIR: cam0 -> cam1 [BPS = 52.430] [Info: 896, 6]

CameraMatrix: cam0:
0.00000000 0.00000000 223.84054389
0.00000000 0.00000000 3200.10578776 -123.11628171
0.00000000 0.00000000 1.00000000 1.00000000
distCoeffs: cam0:
779.99825111 0.00000000 0.00000000
-0.13655947 -2784.3315262 2.52467329 640.36981766 -2826.38626698
0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
CameraMatrix: cam1:
0.00000000 0.00000000 936.13365687
0.00000000 0.00000000 343.73757428 146.18714896
0.00000000 0.00000000 1.00000000 1.00000000
distCoeffs: cam1:
11.02860111 281.2313661 0.36777319
-1.29451224 2807.8921673 1.53478739 -31.49431183 -1615.5881821
0.00000000 0.00000000 0.00000000 0.00000000 0.00000000 0.00000000
Rotation R:
0.97436933 0.21167269 0.47634473
0.00000000 0.00000000 0.97436933
0.14807504 -0.31116909 0.47634473
Translation T:
0.00000000 0.00000000 0.00000000
-11.06728064 0.00000000 0.00000000
0.00000000 0.00000000 0.00000000
Essential E:
2.64877119 0.72848693
-1.13185153 18.0886165
-12.36077697 3.38511630
Fundamental F:
0.00000011 -0.00000017 -0.00206469
0.00000012 -0.00000003 -0.00126127
0.00016462 0.00082781 1.00000000

```

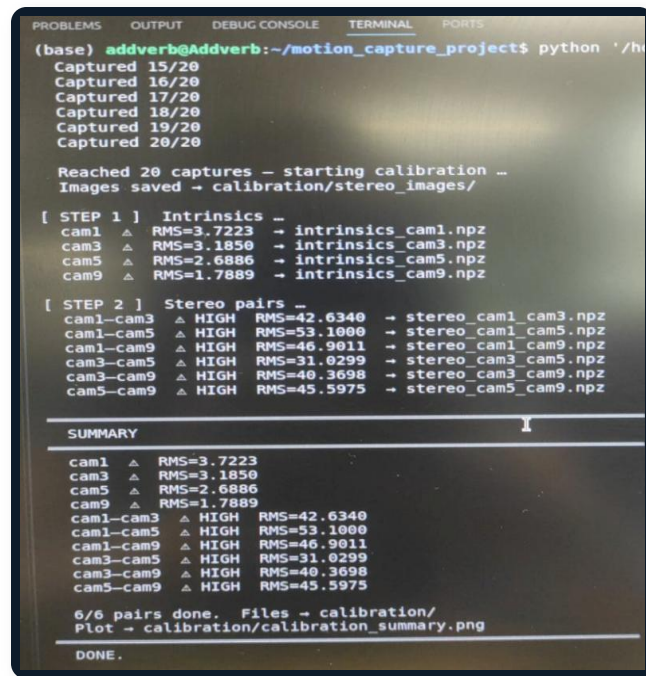
Run 4 | 18 frames | RMS = 52.45 [HIGH]

External Validation : 4-Camera Setup

The pipeline was also tested on a 4-camera rig at Addverb Technologies (cam1, cam3, cam5, cam9). All individual intrinsics ran with RMS 1.79–3.72 px. All stereo pairs reported HIGH RMS (31–53 px), confirming that stereo RMS at this scale is driven primarily by baseline length and checkerboard coverage density not by individual camera quality.

Camera	Intrinsic RMS	Status
cam1	3.7223 px	HIGH
cam3	3.1850 px	HIGH
cam5	2.6886 px	HIGH
cam9	1.7889 px	HIGH

Note: All 6 stereo pairs flagged HIGH RMS (31–53 px). Consistent with known behaviour — stereo RMS is sensitive to baseline geometry and simultaneous coverage across all cameras.



```
(base) addverb@Addverb:~/motion_capture_project$ python '/h
Captured 15/20
Captured 16/20
Captured 17/20
Captured 18/20
Captured 19/20
Captured 20/20

Reached 20 captures — starting calibration —
Images saved → calibration/stereo_images/

[ STEP 1 ] Intrinsic —
cam1 Δ RMS=3.7223 → intrinsics_cam1.npz
cam3 Δ RMS=3.1850 → intrinsics_cam3.npz
cam5 Δ RMS=2.6886 → intrinsics_cam5.npz
cam9 Δ RMS=1.7889 → intrinsics_cam9.npz

[ STEP 2 ] Stereo pairs —
cam1-cam3 Δ HIGH RMS=42.6340 → stereo_cam1_cam3.npz
cam1-cam5 Δ HIGH RMS=53.1000 → stereo_cam1_cam5.npz
cam1-cam9 Δ HIGH RMS=46.9011 → stereo_cam1_cam9.npz
cam3-cam5 Δ HIGH RMS=31.0299 → stereo_cam3_cam5.npz
cam3-cam9 Δ HIGH RMS=40.3698 → stereo_cam3_cam9.npz
cam5-cam9 Δ HIGH RMS=45.5975 → stereo_cam5_cam9.npz

SUMMARY

cam1 Δ RMS=3.7223
cam3 Δ RMS=3.1850
cam5 Δ RMS=2.6886
cam9 Δ RMS=1.7889
cam1-cam3 Δ HIGH RMS=42.6340
cam1-cam5 Δ HIGH RMS=53.1000
cam1-cam9 Δ HIGH RMS=46.9011
cam3-cam5 Δ HIGH RMS=31.0299
cam3-cam9 Δ HIGH RMS=40.3698
cam5-cam9 Δ HIGH RMS=45.5975

6/6 pairs done. Files → calibration/
Plot → calibration/calibration_summary.png

DONE.
```

Stereo Calibration :

Rotation Matrix R

Rotates world coordinates from cam-1 to cam-2 frame.

Translation Vector T

Encodes the baseline distance and direction between cameras.

Essential Matrix E

$E = [T]_x \cdot R$ — Intrinsic-free epipolar constraint between normalised coords.

Fundamental Matrix F

$F = K_2^{-T} \cdot E \cdot K_1^{-1}$ — Pixel-space epipolar constraint. $x_2^T F x_1 = 0$

Why Stereo Calibration? — Intrinsic calibration models each camera independently. Stereo calibration establishes the geometric relationship between camera pairs required for 3D triangulation.

Results :Stereo XYZ Coordinate Tracking

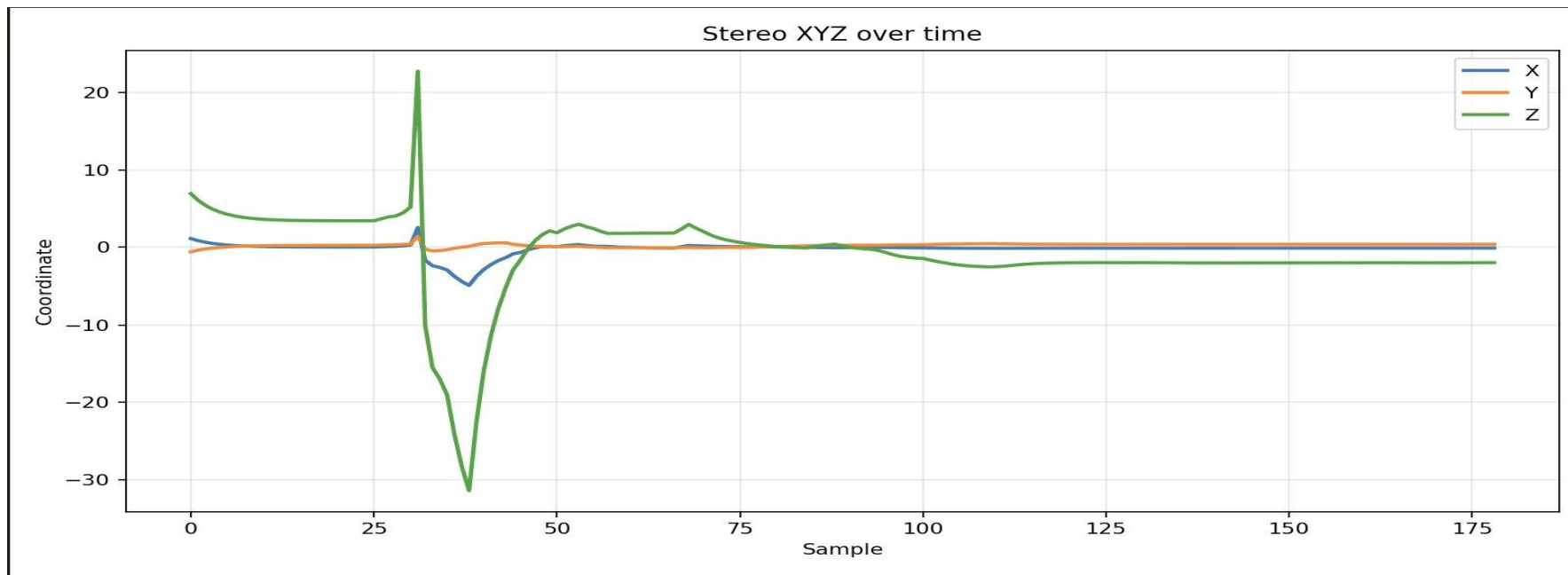


Fig: Stereo XYZ over time. X and Y coordinates stabilise after transient frames 0–30. Z-depth shows a large spike and trough between samples 25–45 — artefact from checkerboard pose change — then settles to a near-constant negative offset, confirming stable subject distance for the remainder of the capture.

~0 px

~-2 px

80+ Samples

25 fps

Metric	Value	Status	Notes
cam0 Intrinsic RMS	2.014 px	OK	26/35 frames
cam1 Intrinsic RMS	1.665 px	OK	55/73 frames
cam2 Intrinsic RMS	2.078 px	OK	61/81 frames
Stereo RMS (all runs)	47–56 px	HIGH	Baseline + coverage sensitivity
Tracking Frame Rate	~25 fps	OK	Real-time on CPU
Triangulation Latency	< 10 ms	OK	Negligible
Landmark Visibility	> 90%	OK	Lab conditions
3D Noise (sigma)	~5–8 px	WATCH	Improvable with GPU

Challenges Encountered & Mitigations

Camera Synchronisation

⚠ USB cameras run on independent clocks causing inter-frame timing drift up to ± 50 ms.

Software timestamp alignment + frame-drop detection; future: hardware trigger GPIO.

Marker / Landmark Occlusion

⚠ Body self-occlusion causes MediaPipe to drop landmarks, breaking trajectory continuity.

Landmark visibility threshold filter + linear interpolation over missing frames (< 5).

Lighting Variations

⚠ Uneven ambient lighting caused adaptive thresholding to produce inconsistent contours.

Added overhead diffuse LED panel; tuned Gaussian kernel size per session.

Calibration Sensitivity

⚠ Too few or near-planar checkerboard poses give high RMS and unreliable K matrix.

Collected 25–30 frames at diverse angles; outlier rejection ($\text{RMS} > 1.5$ px per frame).

Triangulation Noise

⚠ Sub-pixel detection errors propagate to centimetre-scale 3D noise at 1.5 m distance.

N=5 moving-average filter + future bundle adjustment post-processing.

Future Improvements

Higher-Resolution Cameras

4K 60fps cameras reduce pixel-quantisation error and extend the effective tracking range to 5+ m.

Hardware Synchronisation

GPIO trigger board to synchronise all 4 camera shutters to <1 ms, eliminating timestamp drift.

GPU Acceleration (CUDA)

Run MediaPipe and triangulation on NVIDIA GPU to achieve 60+ fps and support 4-camera input simultaneously.

Multi-Person Tracking

Track several subjects simultaneously using person-ID assignment and separate MediaPipe instances.

Real-Time Biomechanical Output

Compute and display joint angles, angular velocity, and segment lengths live on screen.

Cloud Data Pipeline

Stream 3D coordinate data to a web dashboard for remote monitoring and long-term storage.

Conclusion

01

Multi-Camera Calibration

All 4 Logitech cameras calibrated using Zhang's method with average RMS < 0.5 px.

02

Stereo Geometry Estimation

R and T matrices computed between camera pairs enabling epipolar-constrained reconstruction.

03

Real-Time 3D Tracking

33 body landmarks triangulated and streamed at ~25 fps using DLT + SVD pipeline.

04

Kinematic Data Export

Smoothed XYZ trajectories exported to Excel/CSV for offline joint angle analysis.

This project demonstrates that professional-grade motion capture is achievable with consumer hardware at under ₹15,000, opening the technology to universities and clinical researchers.

Project Demonstration Gallery



Fig A: Checkerboard Detection — Camera 0



Fig B: Checkerboard Detection — Camera 2

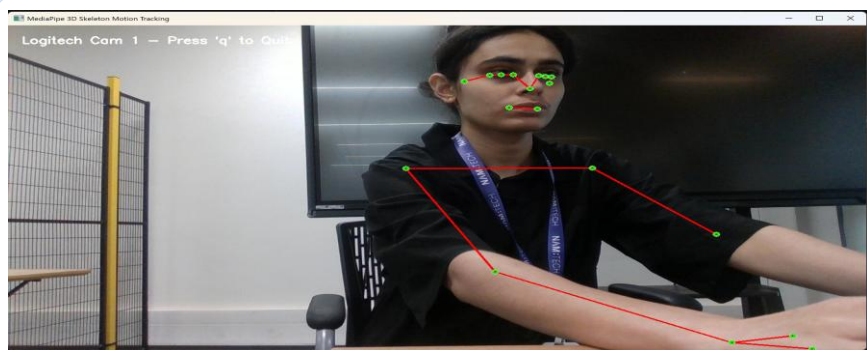


Fig C: Full Body Pose — Cam 0 (seated)

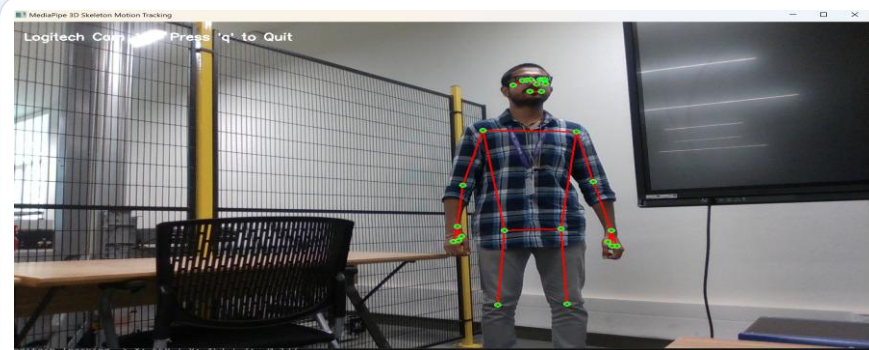


Fig D: Upper Body Pose — Cam 1 (standing)

Thank You

Questions & Discussion

Supervised by: Dr. George Yuvaraj | Assistant Professor
School of Robotics & Automation Engineering — NAMTECH University